

I am on the starting line of the 35 km race walk at the World Championships. I had my usual pre-race meal (carbohydrate based, with a cup of coffee) two hours before the event, along with my sodium bicarbonate loading protocol. This is where I have 20 g of bicarb in capsules (~300 mg/kg) with 1600 of fluid to give me the benefits of a fluid load as well as extra buffering capacity for my race. Then I had a 200 mg dose of caffeine from gum while I was in the call room, waiting to move to the race start. However, an electrical storm has come through and the race has been postponed for 2 hours. What should I do? I'm feeling pretty nervous anyway because this is something I have never encountered before. Do I top up my caffeine and bicarb doses before I start? I usually take another 200 mg caffeine at the 20 km mark of the race but now I don't know if I will have too little or too much on board to execute my race depending on whether I take more now. I have extra supplements available. But how much should I use if decide to top up?

You've found yourself in a unique situation that no study has ever had to consider! Given all the work you've put into your preparation and the importance of the race, it's not surprising that you're feeling rattled. However, you've done a smart thing in calling in some help. In fact, you could reframe the situation as an opportunity rather than a threat. Everyone else in this race is facing these challenges, but you can create an advantage for yourself by making some smart decisions and feeling calm and confident about your pivot. What matters most is making sensible adjustments rather than trying to force the original routine into a new timetable. Let's work through the issues.

The big ticket items for the 35 km race are fuelling and hydration and your pre-race meal has topped up both. The good news is that your muscle glycogen isn't going anywhere. But if you're feeling that you might get hungry or need a blood glucose top up, you could have a small snack now or take a sports gel when you do your new pre-race warm up. That's an easy fix. The sodium hyperhydration protocol can also extend its benefits to the new race start. Again, you might want to sip on some fluid in the warm up so that you're feeling comfortable around hydration. Stay cool during the waiting period. If the storm has reduced the environmental temperature from what you expected, you can adjust your drinking schedule during the race. Rest easy that you're on top of these elements, and you've had plenty of experience over your career in practicing different approaches to fuelling and drinking in race.

Now for the things that you might want to tweak. Caffeine works by reducing your perception of effort and a 200 mg dose is probably about 3 mg/kg for you. I imagine your preparation activities have identified this as an effective dose for the first part of the race. The initial race plan was smart in targeting the opportunity for a competitive lead out, with an additional boost when the race pace starts to identify the podium contenders. Although you have already taken your first dose, the half life of caffeine

is generally around 4-6 hours, so you should still be feeling the benefits at the new start time. Your past practices with race caffeine use, and your general experience as a caffeine consumer give you plenty of insights into how you might respond. Too much caffeine can create jitters so the fact that you're already feeling apprehensive is a message that a further pre-race top up could tip you over the edge. The great news is that your chosen caffeine source is a quick acting gum - this gives you options to fine tune your in-race strategy with a rapid fire outcome. Depending on how you are feeling in the early part of the race, you could decide to stick to the 20 km top up, or decide to bring it on early. Gum is an easy thing to carry and to manage during the race, so taking an extra dose with you might even allow for a further boost in the late stages. Lean into the expertise you have developed over your racing career.

The use of bicarbonate in endurance events is a more recent addition to the sports nutrition tool box. Its original use to buffer the excess hydrogen ions produced during sustained efforts at maximal intensity has been shifted to accommodate the race winning strategies within longer events. Since this is a new area, there is little published research and most of the knowledge comes from individual experimentation and studies that are mid implementation. We've tapped into our network to find someone who's at the forefront of this work. You've probably heard of Dr Trent Stellingwerff – coach, former athlete and expert sports scientist. His wisdom in this area is here to help.

Trent says it's great that you have invested in this emerging area. Your experience and preparation are your biggest asset and can be easily tapped into to fully exploit the benefits of the bicarb loading protocol. Your original strategy probably made use of knowledge about your individual time to peak blood bicarbonate concentrations, and no doubt you had race tactics involving a surge in pace at critical times to allow you to exploit your enhance buffering capacity. You may not have appreciated that your blood bicarbonate levels will stay elevated for many hours, and it is the physiology of your race tactics that will likely deplete them rather than a decay with time. This should give you confidence to execute your pace plan as intended – whether that was to cover surges within the leader pack, or to launch your own pace increase at a critical time. There is likely little benefit – and potentially a risk of side effects – from taking a top up. Your race preparation has determined a dosing schedule that is both effective and with minimal risk of the well-known side effects of bicarb loading such as gastrointestinal problems. Have confidence that it had inbuilt flexibility to cover the unforeseen weather events today. Stick to the plan with renewed belief that your due diligence and preparation have also provided you with resilience. Your fellow competitors may not have prepared with your thoroughness nor invested in having spare expertise on hand. Instead of second guessing your plan or adjusting it in fear, enjoy getting back into your focussed race mode and execute it with the confidence it deserves. You've got this!